BEING NOWHERE DENSE IS A LOCAL PROPERTY ON AN OPEN COVER

THE STACKS PROJECT AUTHORS

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Lemma 0.1. *Let X be a topological space. Let $X = \bigcup U_i$ be an open covering. Let $T \subset X$ be a subset. If $T \cap U_i$ is nowhere dense in U_i for all i, then T is nowhere dense in X.*

Proof. Denote $\overline{T}_i$ the closure of $T \cap U_i$ in U_i. We have $\overline{T} \cap U_i = \overline{T}_i$. Taking the interior commutes with intersection with opens, thus

$$(\text{interior of } \overline{T}) \cap U_i = \text{interior of } (\overline{T} \cap U_i) = \text{interior in } U_i \text{ of } \overline{T}_i$$

By assumption the last of these is empty. Hence T is nowhere dense in X. □

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